



Requirement Analysis Specification Document (RASD)

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1. Introduction

1.1 Purpose

1.1.1 General purpose

SafeStreets is a crowd-sourced application that lets users notify authorities when traffic violations occur. It offers a basic service and two advanced functions. The service can be used upon registration.

Basic service

SafeStreets collects the data (photo, date, time, position and type of the violation) provided by the user, completes it with suitable metadata, processes the picture through an image recognition algorithm and eventually stores the data in its servers (if no errors occurred).

Furthermore, SafeStreets offers its users the possibility to mine the collected information, providing different levels of visibility based on the role of the applicant (citizen or authority).

Advanced Function 1

SafeStreets crosses its own data with those provided by municipalities in order to identify potentially risky areas and suggest possible interventions.

Advanced Function 2

Given the fact that municipalities could use SafeStreets information to issue violation tickets, the application implements a mechanism which ensures data to be always valid, never damaged nor altered. Additionally, SafeStreets uses the information about issued tickets to build statistics or to strengthen the validity of its initiative.

1.1.2 Goals

These are the application goals:

- G1.** The application must allow the user to send photos about parking violations
- G2.** The application must allow the user to consult his own data
- G3.** The application must allow the user and authorities to extract the data about about parking violations

- G4.** The application must provide different levels of visibility of the requested data based on the role of the issuer
- G5.** The application must be capable of retrieving information from municipality (those offering the services) about violations.
- G6.** The application must provide the municipality suggestions about possible interventions to violations
- G7.** The application must allow the municipality to retrieve the information about the violation to generate traffic tickets.
- G8.** The application must be capable of retrieving information from municipality about issued tickets.

1.2 Scope - non so che scrivere

SafeStreets intends to provide users with the possibility to notify authorities when traffic violations occur, and in particular parking violations. It also aims to help authorities to improve their penalty system and get informations about correlations in violations and city's areas.

1.3 Definitions, acronyms, abbreviations

1.3.1 Definitions

1. Autorità = municipality
2. User
- 3.

1.3.2 Acronyms

1.3.3 Abbreviations

1.4 Revision History

- A. 9/11/19 - Version 1.0:
First release;

1.5 References

1. Specification document "SafeStreets Mandatory Project Assignment"
2. IEEE Std 830-1998 IEEE REcommended Practice for Software Requirements Specifications
3. General Data Protection Regulation (GDPR)

1.6 Document Overview

The RASD has the following structure:

Chapter 1 contains a brief introduction to the application. Furthermore it lists the goals which the application has to reach, the scope of the project and some more information needed in order to understand the application;

Chapter 2

Chapter 3

Chapter 4 contains a modelling of the project through the Alloy language;

Chapter 5 lists the number of hours spent by each member on the project.

2.4 Assumptions, dependencies and constraints

Scritte da Francesco

D1: Each license plate is an unique identifier.

D2:The devices that acquire violations' position provide location with an error of 5

meters at most.

D3: The internet connection works properly without failure.

D4: The information about the accidents of the municipality are correct.

D5: The authorities and in particular the municipality have demonstrated that they are acknowledged public institutions.

D6: Municipality is a particular case of authority.

D7: The information coming from the users is never broken if the image is not manipulated and the license plate is readable.

D8: Each municipality that shares information with SafeStreet has authorized the app to take the informations that has inside.

D9: The data and the time is setted correctly on the device of the user and authorities.

D10: System is able to discard non-compliant photos

D11: Users can correctly identify the type of violation

Functional Requirements

User

Scenarios

Scenario 1

Una mattina Alice sta per andare a scuola con la bici e usa come sempre le piste ciclabili della sua città. Ad un certo punto durante il tragitto si trova la strada sbarrata da una macchina parcheggiata proprio sopra la pista ciclabile.

La ragazza a questo punto prende il suo device con l'app di safestreets installata, scatta una foto alla targa dell'auto e poi tramite l'app la invia, inserendo anche tutte le informazioni necessarie, per segnalare la violazione di parcheggio.

A questo punto lei può vedere nello storico della sua app la segnalazione appena inviata nella speranza che possa essere utile per le autorità della città che collaborano con safeStreets.

Scenario 1 - ENG

One day Alice is going to school by bike on the bike path as always. At some point the road is barred by a car parked exactly on the bike path.

The girl decides to use her device using *SafeStreets*, an app installed on it. She takes a photo to the license plate, she adds all the necessary informations needed to report that violation and then she sends the pic and the informations via app.

After that she is now able see in the application's archive her reporting just sent, hoping it could be able to city's authorities which collaborate with *SafeStreets*.

Scenario 2

Durante ferragosto Mario è appena arrivato a Bari e vuole parcheggiare la sua macchina in un luogo dove non incontrerà probabilmente difficoltà di parcheggio.

Per avere informazioni decide di usare l'app di SafeStreets chiedendo informazioni sulla città di Bari

e l'app risponde facendo vedere a Mario le zone di Bari con la più alta frequenza di violazioni di parcheggio e macchine in doppia fila. In questo modo Mario ha una lista delle zone da evitare. L'app non mostra però tutti i dettagli riguardanti i tipi di veicoli e le targhe di essi che hanno fatto la violazione.

Scenario 2 - ENG

In August Mario arrives in Bari and wants to find an area where he won't struggle to park. He decides to use *safestreets* to get this information thinking that a higher rate of violations is associated to a lack of parkings. He looks for info about Bari and he discovers what are Bari's areas with the highest parking violations frequency and the most double-parked cars. Thanks to this Mario knows which zones he has to avoid. The application doesn't show all the details about vehicles and license plates connected to the violations.

Use Case Diagram

Azioni coinvolte:

1. sign up - initiate
2. login - include
3. report - initiate
4. personal data visualization - initiate
5. data mining - participate

Name	Sign Up
Actor	User
Entry Condition	The user has downloaded and opened the application
Events flow	1. The user taps on the sign up option 2. The user fills all the fields (all mandatory) 3. The user confirms his data
Exit conditions	The user confirms his account through email link and system stores his data
Exceptions	1. The user hasn't filled all the fields 2. The user provided wrong data (fiscal code not matching other data, invalid email) 3. The user inserts an already taken username 4. The user password doesn't meet the security requests

Name	Login
Actor	User
Entry Condition	1. The user has downloaded and opened the app. 2. The user has already done the "Sign Up" activity.
Events flow	1. The user chooses the "Login" option. 2. The user enters username and password in the respective fields. 3. The user chooses the confirmation option.

Exit conditions	The user is logged in and he can execute other actions
Exceptions	1. The user enters the wrong username 2. The user enters the wrong password In both cases the user is warned by the application about what is the wrong field and receive the suggestion to update it and retry.

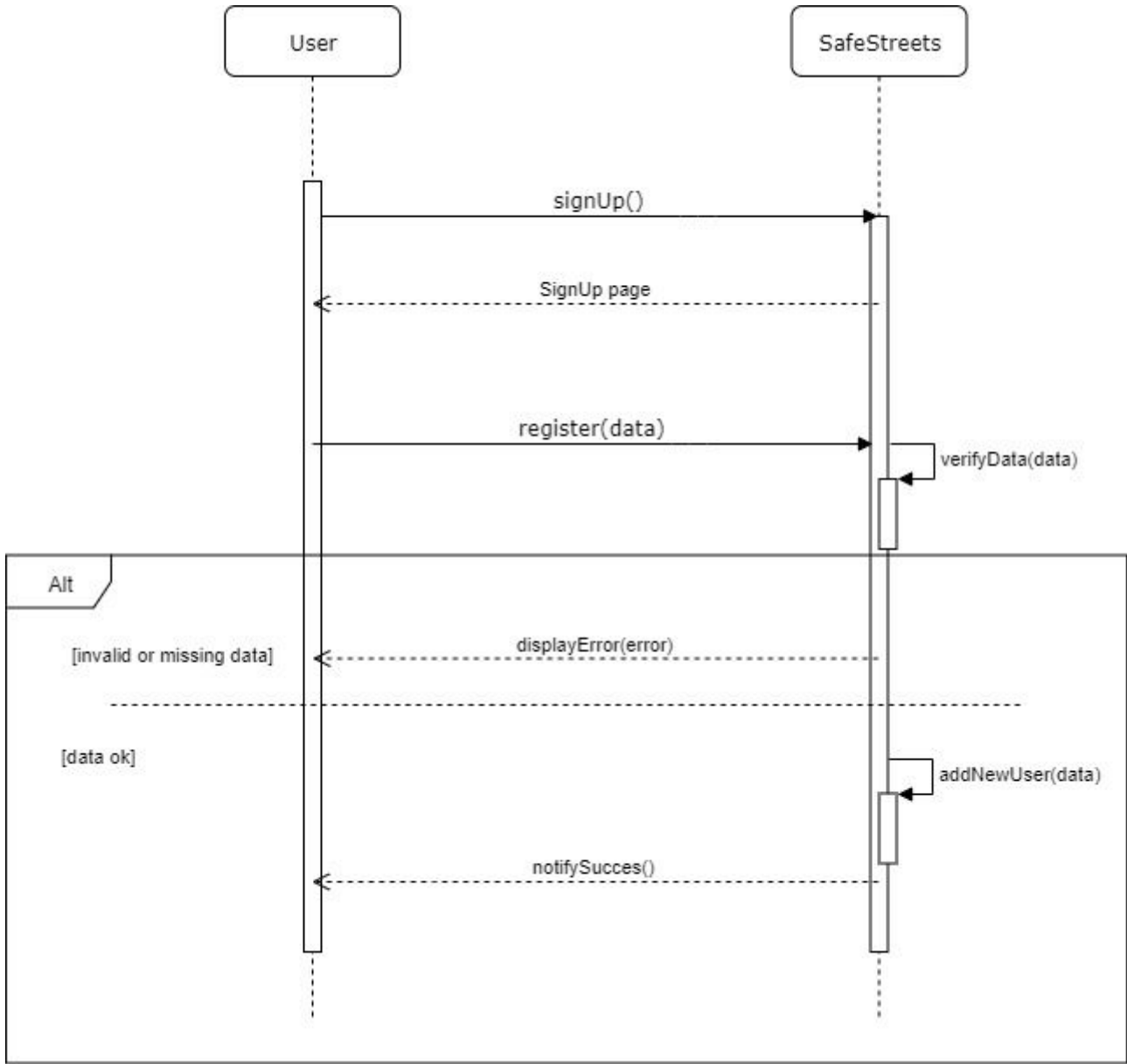
Name	Report
Actor	User
Entry Condition	1. The user has already done the “Login” activity.
Events flow	1. The user choose the “Report violation” option. 2. The user shots a pic of the violation 3. The user adds optional informations to the picture. 4. The system adds technical informations to the pic (e.g. data, time, position, ..) 5. The user sends the report
Exit conditions	1. SafeStreets receives the report
Exceptions	1. GPS is not available. 2. Internet connection is not available In both cases the user is notified about the issue.

Name	Personal data visualization
Actor	User
Entry Condition	1. The user has already done the “Login” activity.
Events flow	1. The user taps on “My reports” 2. The user chooses among the available sent reports 3. The user visualizes his data
Exit conditions	The user taps on close button
Exceptions	1. No available reports

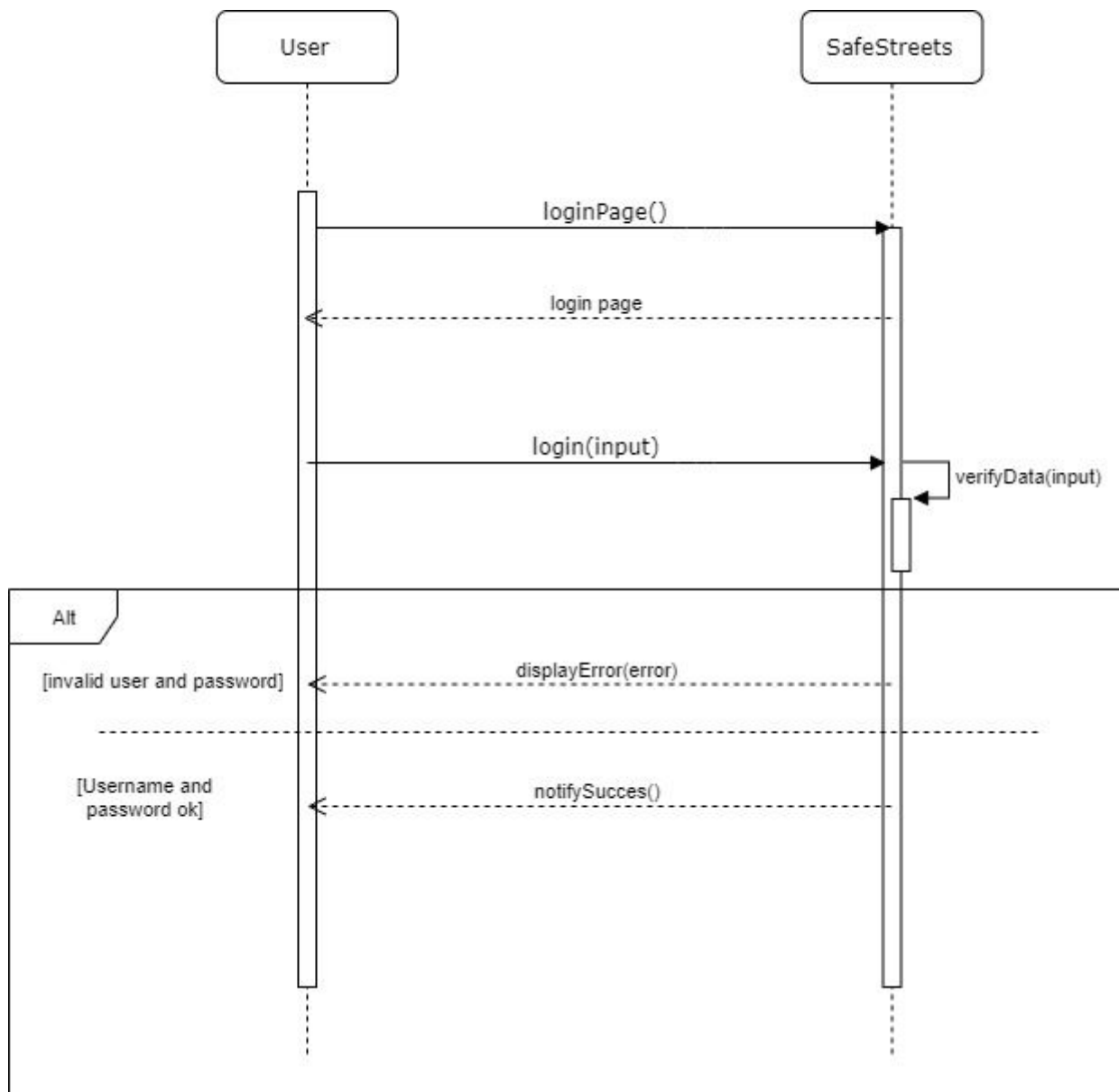
Name	Data mining
Actor	User, other users
Entry Condition	1. The user has already done the “Login” activity.
Events flow	1. The user chooses the “Data mining” option. 2. The user select an area of interest. 3. The information about that area is shown on the

	map through different highlightings.
Exit conditions	The user taps on close button
Exceptions	1. No available data 2. Invalid query

User sign-up



User login



Authority

Scenarios

Scenario 3

Una pattuglia dei vigili urbani vuole pattugliare alcune zone della città di Terni ma vuole capire quali conviene visitare per avere un'alta probabilità di trovare violazioni di parcheggio.

Per fare questo decidono di collegarsi a *SafeStreets* con l'account della polizia locale e chiedono informazioni sulla città di Terni. Il servizio risponde facendo vedere le zone di Terni con la più alta frequenza di violazioni di parcheggio e inoltre fa vedere anche le tipologie di veicoli che hanno commesso più violazioni le targhe delle violazioni più recenti.

Scenario 3 - ENG

A police's unit has to patrol some areas of Terni and needs to understand where is more probably to find parking violations. Thus, the unit decides to connect to *SafeStreets* using the police's account and looks for informations about Terni. The policemen using the service can see which are the Terni's areas

connected to an higher probability of parking violations and they can also read not only what kind of vehicles committed violations often but also the license plates of the more recents infractions.

Scenario 4

Il comune di Milano riceve periodicamente report da *SafeStreets* indicanti il numero di auto segnalate da utenti coinvolte in incidenti nelle zone del centro città. *SafeStreets* suggerisce la riduzione del limite di velocità in queste aree. Il comune, accortosi della vicinanza di queste zone ad aree frequentate da bambini e ragazzi (scuole, parchi), raccoglie il suggerimento di *SafeStreets* e inoltre procede all'installazione di dossi rallentatori.

The municipality of Milan periodically receives reports from *SafeStreets* indicating how many user-reported cars were involved in traffic accidents in the city centre. *SafeStreets* suggests a reduction of the speed limit in those areas. The Municipality, given the proximity of that area to places frequented by children (schools, parks), decides to execute the suggestion and to install some speed bumps.

SafeStreet ha appena ricevuto autorizzazione dal comune di Milano per poter prendere dati sugli incidenti avvenuti nel territorio. Il comune ha appena registrato il verificarsi di un incidente in Corso Sempione. *SafeStreet* prende le informazioni sull'incidente dal comune, poi incrocia queste informazioni con le sue per segnare le aree non sicure e il comune in risposta riceve una serie di suggerimenti da attuare durante l'intervento da parte delle forze dell'ordine per mettere l'area dell'incidente al sicuro per poterne evitare altri.

Scenario 5

Il comune di Genova decide di affidarsi al servizio *SafeStreets* per poter automatizzare il sistema di generazione multe nelle zone periferiche della città, raramente presidiate dalle pattuglie. Il comune, dopo essersi iscritto al servizio, riceve periodicamente le segnalazioni della zona fatte dagli utenti e le inoltra a una divisione della municipale che, da casa, analizza e convalida le segnalazioni. Una volta generate le multe, la Municipalità invia le informazioni sulle multe emesse a *SafeStreets*. *SafeStreets* prende le informazioni e aggiorna le sue statistiche che misurano l'efficienza del servizio.

Scenario 5 - ENG

The municipality of Genova decides to buy *SafeStreets* service so that it can automate its penalty system for the city suburbs, barely supported by patrols. The municipality, once subscribed to the service, periodically receives the user made reports and forwards them to a special division which reviews and convalidates such reports, from home. Once the municipality has issued the violation tickets, it notifies *SafeStreets*. *SafeStreets* collects the information and updates its statistics about the efficiency of the service.

Use Case Diagram

Azioni coinvolte:

1. Sign up
2. login
3. data mining
4. receive suggestions
5. receive reports

Name	Sign Up
Actor	Authority
Entry Condition	The authority has downloaded and opened the application
Events flow	1. The authority taps on the sign up option 2. The authority fills all the fields (all mandatory) 3. The authority confirms his data
Exit conditions	The authority confirms his account through email link and system stores his data
Exceptions	1. The authority hasn't filled all the fields 2. The authority provided wrong data (fiscal code not matching other data, invalid email) 3. The authority inserts an already taken username 4. The authority password doesn't meet the security requests

Name	Login
Actor	Authority
Entry Condition	1. The authority has downloaded and opened the app. 2. The authority has already done the "Sign Up" activity.
Events flow	1. The authority chooses the "Login" option. 2. The authority enters username and password in the respective fields. 3. The authority chooses the confirmation option.
Exit conditions	The authority is logged in and he can execute other actions
Exceptions	1. The authority enters the wrong username 2. The authority enters the wrong password In both cases the authority is warned by the application about what is the wrong field and receive the suggestion to update it and retry.

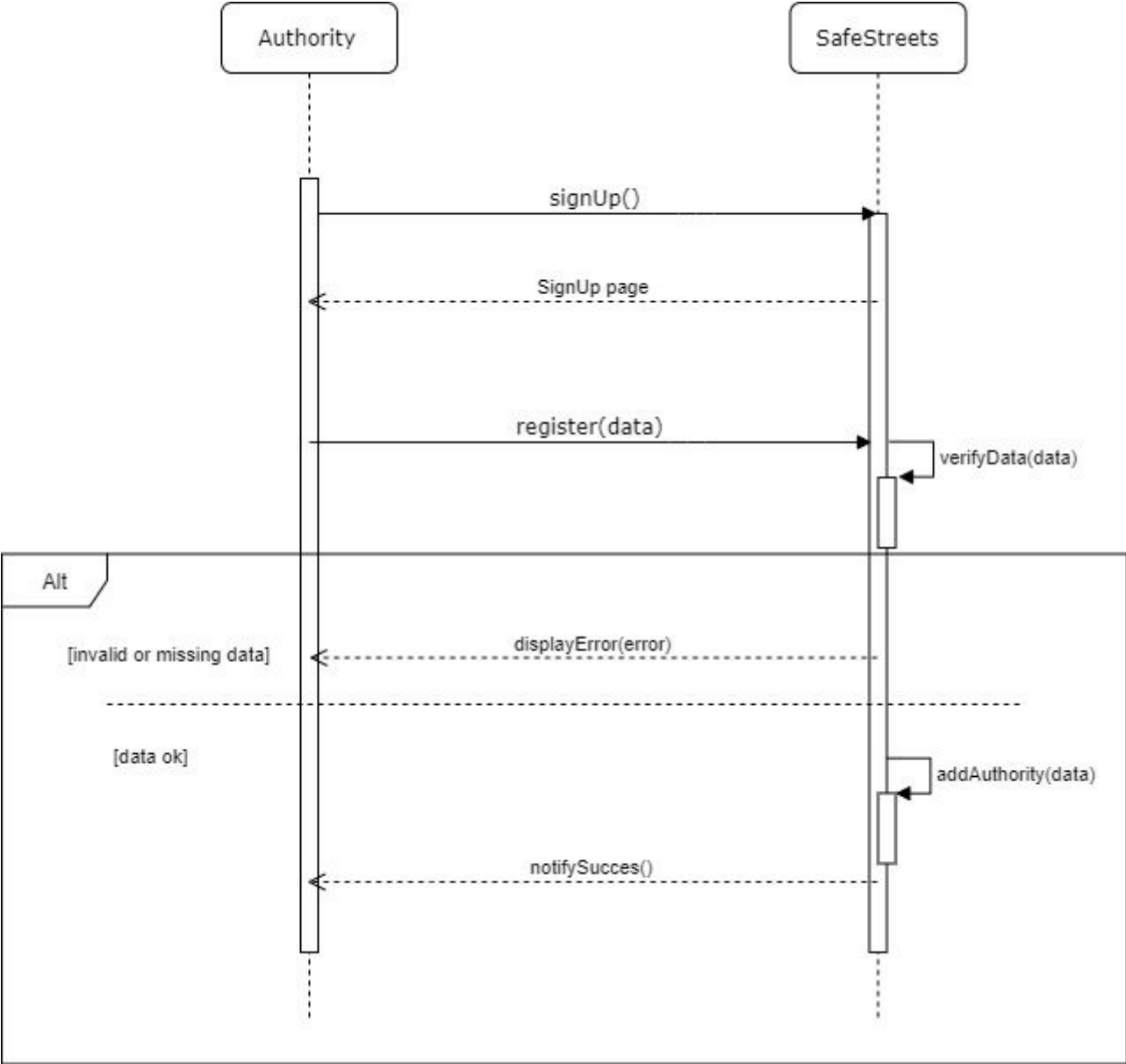
Name	Data mining
Actor	Authority, other users
Entry Condition	1. The authority has already done the "Login" activity.
Events flow	1. The authority chooses the "Data mining" option. 2. The authority select an area of interest. 3. The information about that area is shown on the map through different highlightings.

Exit conditions	The authority taps on close button
Exceptions	1. No available data 2. Invalid query

Name	Receive suggestions
Actor	Authority
Entry Condition	1. The authority has already done the “Sign up” activity. 2. The authority has subscribed to the <i>SafeStreets</i> service. 3. <i>SafeStreets</i> has performed an analysis about a specific area.
Events flow	1. The authority is notified with the informations
Exit conditions	The authority receives the suggestions
Exceptions	1. Not enough data to perform analysis

Name	Receive reports
Actor	Authority, other users
Entry Condition	1. A registered user uses the service for reporting a violation 2. The authority has already done the “Sign up” activity
Events flow	1. The user sends the report. 2. <i>SafeStreets</i> analyzes the report verifying it is not forged. 3. The report is delivered to the authority. 4. The authority selects the “Segnalations” option in the menù. 5. The authority can see and analyze the report.
Exit conditions	The authority taps on close button.
Exceptions	1. The report is forged.

Authority SignUp



Authority Login

